FUNCTIONAL IMPROVEMENT

MOVING BEYOND DYSTROPHIN IN DMD



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Disclosures

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Unmet needs in DMD

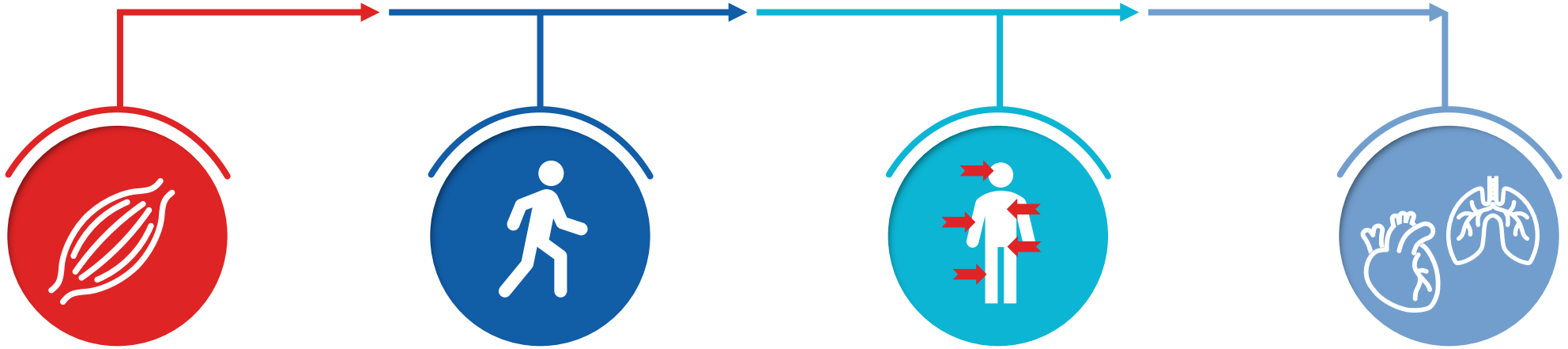
Aravindhan Veerapandiyan, MD



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DMD is caused by dystrophin deficiency and manifests as a progressive multisystem disease, affecting muscle and the CNS

Greatly reduced or absent dystrophin protein expression[1]



Muscle weakness and gait abnormalities[2]

Progressive loss of upper and lower limb strength and function[2]

Multisystem involvement that affects skeletal, including pulmonary, smooth, and cardiac muscles and the CNS[3]

Cardiopulmonary failure as the leading cause of death[4]

CNS, central nervous system; DMD, Duchenne muscular dystrophy.

1. de Feraudy Y, et al. *Ann Neurol.* 2021;89:280–292; 2. Bushby K, et al. *Lancet Neurol.* 2010;9:77–93; 3. Ohlendieck K, Swandulla D. *Pflügers Arch.* 2022;473:1813–1839; 4. Birnkrant DJ, et al. *Lancet Neurol.* 2018;17:251–267.

Therapeutic approaches in DMD have evolved from supportive care to targeted approaches

Supportive standard of care[1–3]

~1990s[2]



- Corticosteroid therapy[1,2]
- Multidisciplinary management[3,4]

Dystrophin-producing therapies[1,2]

2016*



Exon skipping therapy[2]

- RNA-level dystrophin restoration



Gene therapy[1]

- Micro-dystrophin protein replacement

HDAC inhibitors[1]

2024[1]



- Target inflammation, fibrosis, muscle degeneration[5]

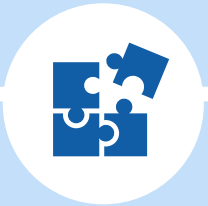
DMD, Duchenne muscular dystrophy; HDAC, histone deacetylase.
*The first exon skipping therapy was approved in 2016[2]; the first gene therapy was approved in 2023[1].

1. Komaki H. *Brain Dev*. 2025;47:104397; 2. D'Ambrosio ES, Mendell JR. *Neurotherapeutics*. 2023;20:1669–1681; 3. Bushby K, et al. *Lancet Neurol*. 2010;9:177–189; 4. Duan D, et al. *Nat Rev Dis Primers*. 2021;7:13; 5. Anjum AF, et al. *Curr Ther Res Clin Exp*. 2025;102:100787.

Despite progress, challenges remain with current treatments

Exon skipping and gene therapy

Dystrophin quantity and quality



- Low dystrophin expression levels[1]
  - <1% with approved exon 51 skipping therapy
- Micro-dystrophin functionality[1]
  - Shorter micro-dystrophin protein lacks key functional domains believed to support muscle health[2,3]

Effective distribution and durability of effect



- Limited delivery to muscle and the CNS[1,4]
- Modest clinical benefits with exon skipping[4]
- Unknown durability of effect with gene therapy[4]

All modalities

Treatment burden, safety, and access

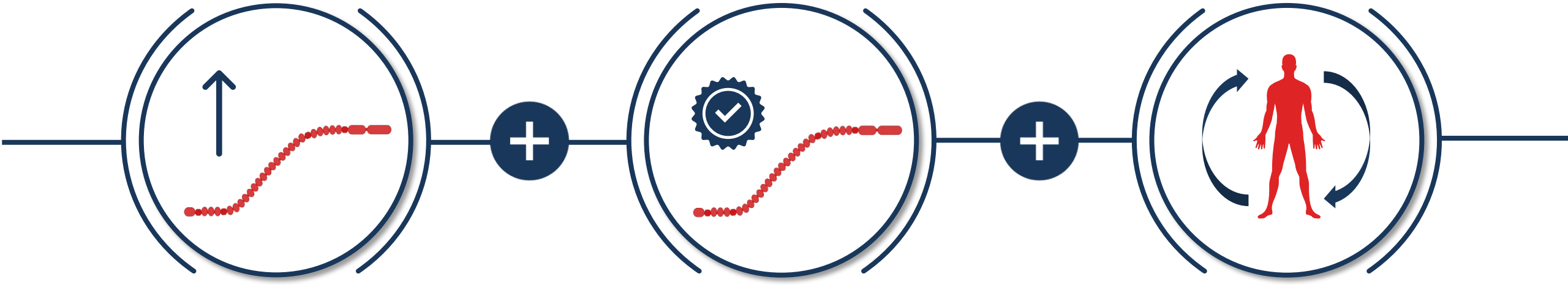


- Access/clinical eligibility restrictions with gene therapy[4]
- Adverse events[1,5,6]
- Regular monitoring[5,7]
- Dosing
  - Inability to safely and effectively re-dose with gene therapy[1,8]
  - Frequent IV dosing with current exon skipping therapy[8]

CNS, central nervous system; IV, intravenous.

1. Chwalenia K, et al. *J Muscle Res Cell Motil*. 2025;46:293–300; 2. Harper S, et al. *Hum Mol Genet*. 2002;11:1807–1815; 3. Davies KE, Guiraud S. *Mol Ther*. 2019;27:486–488; 4. Gonzalez Castillo Z, et al. *J Transl Genet Genom*. 2025;9:338–351; 5. Montagna C, et al. *Int J Mol Sci*. 2025;26(:6742; 6. D'Ambrosio ES, Mendell JR. *Neurotherapeutics*. 2023;20:1669–1681; 7. Das P, et al. *Pharmacology*. 2025;110:1–13; 8. Komaki H. *Brain Dev*. 2025;47:104397.

Functional improvement in DMD requires therapeutic approaches that improve the quantity, quality, and distribution of dystrophin



Increased dystrophin production

Production of high-quality dystrophin

Broad and effective distribution of dystrophin restorative treatments to affected muscles and the CNS

Functional improvement

CNS, central nervous system; DMD, Duchenne muscular dystrophy.



Even small, measurable changes – an extra few steps, an easier rise, a longer walk, retaining upper arm strength – can make a real difference for patients and families

THANK YOU



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